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Devices, systems and methods for transmitting motion to the distal tip of an endoscopic accessory tool

Abstract

The present disclosure relates generally to the field of medical devices and establishing access to body passageways. In particular, the present disclosure relates to devices, systems and methods to facilitate entry of an endoscopic accessory tool into and/or through patient-specific anatomies. For example, the devices, systems and methods of the present disclosure may transmit mechanical motion to the distal end of an endoscopic accessory tool to facilitate atraumatic access to tortuous or otherwise restricted anatomies.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of and claims the benefit of the earlier filing date of U.S. patent application Ser. No. 16/242,416, filed on Jan. 8, 2019, which claims the benefit of priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application No. 62/615,109, filed on Jan. 9, 2018, and which applications are incorporated herein by reference in their entireties for all purposes.

FIELD

(1) The present disclosure relates generally to the field of medical devices and establishing access to body passageways. In particular, the present disclosure relates to devices, systems and methods to facilitate entry of an endoscopic accessory tool into and/or through patient-specific anatomies.

BACKGROUND

(2) Medical professionals face a variety of technical challenges when performing endoscopic cannulation procedures that require advancing a guidewire and/or endoscopic accessory tool (e.g., sphincterotome, cannula, catheter, etc.) into and through challenging patient anatomies. For example, a target body passageway may be oriented at a difficult angle relative to the endoscopic accessory tool, have a very small or sealed opening, or include a tortuous anatomy, blockages (e.g., stones, etc.) and benign or malignant structures. To facilitate smooth and efficient entry of a guidewire and endoscopic accessory tool into/through a target body passageway, medical professionals often manually rotate, oscillate, linearly advance and/or reciprocate the endoscopic accessory tool, and by proxy the guidewire itself, to “wiggle” the guidewire and endoscopic accessory tool and thereby reduce friction exerted against the opening and/or inner wall of the passageway. As might be expected, precise control of the force and frequency of movement imparted to the accessory tool and guidewire largely remains crude and uncontrolled. Even experienced medical professionals often require multiple attempts to achieve successful cannulation, especially when working against the natural friction and abnormal patient-specific pathologies of the specific body passageway. Naturally, the likelihood of causing trauma to the tissues comprising or surrounding the target body passageway increases with the number of cannulation attempts. In some instances, the medical professional may be required to abort the cannulation procedure entirely. In other instances, the traumatized tissues may be prone to harmful and potentially life-threatening post-operative inflammation.

(3) A variety of advantageous medical outcomes may be realized by the devices, systems and methods of the present disclosure which facilitate enhanced entry of a guidewire and endoscopic accessory tool into and/or through challenging patient-specific anatomies.

SUMMARY

(4) In one aspect, the present disclosure relates to a medical device comprising an elongate tubular body having a proximal end, a distal end and a length therebetween. The elongate tubular body may include a first lumen extending between the proximal and distal end. A distal portion of the elongate tubular body may include an opening in communication with the first lumen. An elevator may be disposed at a first location within the distal portion of the elongate tubular body. The elevator may be configured to move between a first position and a second position relative to a longitudinal axis of the elongate tubular body. An actuator may be disposed at a second location within the distal portion of the elongate tubular body. The second location may be adjacent to the first location such that the actuator may be configured to transmit mechanical motion to the

elevator. The second location may be on one side of the first lumen proximal to the elevator. Alternatively, the second location may be on opposite sides of the first lumen proximal to the elevator. The elongate tubular body may further include a second lumen adjacent to first lumen. The first lumen may be configured to receive an endoscopic accessory tool therethrough. The elevator may be configured to act upon a distal portion of the endoscopic accessory tool as the elevator moves between the first and second positions. The actuator may be configured to transfer mechanical motion to the distal portion of the endoscopic accessory tool.

(5) In another aspect, the present disclosure relates to a medical device comprising an elongate tubular body having a proximal end, a distal end, and a length therebetween. The elongate tubular body may include a first lumen extending between the proximal and distal ends. A distal portion of the elongate tubular body may include an opening in communication with the first lumen. An elevator may be disposed at a first location within a distal portion of the elongate tubular body. The elevator may be configured to move between a first position and a second position relative to a longitudinal axis of the elongate tubular body. An endcap may be disposed around the distal portion of the elongate tubular body. The endcap may include an opening configured to align with the opening of the elongate tubular body. An actuator may be disposed within the endcap adjacent to the opening of the endcap. The actuator may be configured to transmit mechanical motion across the opening of the endcap. The first lumen may be configured to receive an endoscopic accessory tool therethrough. The elevator may be configured to act upon a distal portion of the endoscopic accessory tool as the elevator moves between the first and second positions. The actuator may be configured to transmit mechanical motion to a portion of the endoscopic accessory tool extending through the opening of the endcap.

(6) In yet another aspect, the present disclosure relates to an endoscopic accessory tool comprising an elongate member having a proximal portion with a proximal end, a distal portion with a distal end and a lumen extending therebetween. The distal portion of the elongate member may be configured to receive energy propagated along a length of the elongate member to transmit motion to the distal portion of the elongate member. The motion may be transmitted in a plurality of directions relative to a longitudinal axis of the elongate member. The lumen may be configured to receive a guidewire therethrough. An actuator may be disposed on or within a portion of the elongate member. The actuator may be attached to the distal end of the elongate member. The actuator may be disposed within the proximal portion of the elongate member. The energy may include vibrational energy, and the actuator may propagate the vibrational energy along the length of the elongate member to transmit mechanical motion to the distal portion of the elongate member. A cutting wire may extend along the length of the elongate member. A proximal end of the cutting wire may be connected to an energy source. The energy source may be configured to deliver cauterizing energy to a distal portion of the cutting wire. A portion of an inner surface of the elongate member may include a surface feature configured to slidably engage a corresponding surface feature on an outer surface of the guidewire. The surface features may be disposed on the proximal portion of the elongate member and a proximal portion of the guidewire. The surface features may be disposed on the distal portion of the elongate member and a distal portion of the guidewire. The surface feature of the guidewire may extend around a full circumference of the guidewire. The surface feature of the guidewire may extend around less than a full circumference of the guidewire. The elongate member may be configured to extend through a medical device. The medical device may include an elongate tubular body having a proximal end, a distal end and a length therebetween. The elongate tubular body may include a first lumen extending between the proximal and distal ends. A distal portion of the elongate tubular body may include an opening in communication with the first lumen. An elevator may be disposed at a first location within the distal portion of the elongate tubular body. The elevator may be configured to move between a first position and a second position relative to a longitudinal axis of the elongate tubular body. An actuator may be disposed at a second location within the distal portion of the elongate tubular body.

The actuator of the medical device may be configured to transfer mechanical motion to the distal portion of the endoscopic accessory tool. In various embodiments, the actuator of the medical device may be configured to transfer mechanical motion to the distal portion of the endoscopic accessory tool in addition to, or as an alternative to, the actuator disposed on or within a portion of the elongate member.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Non-limiting embodiments of the present disclosure are described by way of example with reference to the accompanying figures, which are schematic and not intended to be drawn to scale. In the figures, each identical or nearly identical component illustrated is typically represented by a single numeral. For purposes of clarity, not every component is labeled in every figure, nor is every component of each embodiment shown where illustration is not necessary to allow those of ordinary skill in the art to understand the disclosure. In the figures:
- (2) FIG. 1 provides a schematic illustration of a medical device disposed within a body passageway of a patient, according to one embodiment of the present disclosure.
- (3) FIGS. 2A-2C provide perspective views of a distal portion of an endoscopic accessory tool, according to one embodiment of the present disclosure.
- (4) FIGS. 3A-3D provide perspective side views of a medical device and endoscopic accessory tool, according to one embodiment of the present disclosure.
- (5) FIG. 4 provides a perspective side view of a medical device and endoscopic accessory tool, according to one embodiment of the present disclosure.
- (6) FIGS. 5A-5D provide perspective side (FIGS. 5A and 5C) and front (FIGS. 5B and 5D) views of a distal portion of an endoscopic accessory tool, according to one embodiment of the present disclosure.
- (7) FIGS. 6A-6B provide perspective views of an endoscopic accessory tool, according to one embodiment of the present disclosure.
- (8) FIGS. 7A-7D provide perspective views of an endoscopic accessory tool and guidewire, according to one embodiment of the present disclosure.
- (9) FIGS. 8A-8D provide perspective views of an endoscopic accessory tool and guidewire, according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

- (10) The present disclosure is not limited to the particular embodiments described. The terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting beyond the scope of the appended claims. Unless otherwise defined, all technical terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure belongs.
- (11) Although embodiments of the present disclosure are described with specific reference to medical devices and systems (e.g., endoscopic accessory tools and/or guidewires inserted through a duodenoscope, etc.) for selective cannulation of the common bile duct (CBD) or pancreatic duct (PD) during an Endoscopic Retrograde Cholangiopancreatography (ERCP) procedure, it should be appreciated that such medical devices and systems may be used in a variety of medical procedures which require navigating one or more accessory tools through ductal, luminal, or vascular anatomies, including, for example, interventional radiology procedures, balloon angioplasty procedures, thrombolysis procedures, angiography procedures and the like. The disclosed medical devices and systems may be inserted via different access points and approaches, e.g., percutaneously, endoscopically, laparoscopically or some combination thereof.
- (12) As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms

as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used herein, specify the presence of stated features, regions, steps, elements and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components and/or groups thereof.

(13) As used herein, the term “distal” refers to the end farthest away from the medical professional when introducing a device into a patient, while the term “proximal” refers to the end closest to the medical professional when introducing a device into a patient.

(14) Selective cannulation refers to the act of passing a guidewire and endoscopic accessory tool through the major papilla **4** (e.g., ampullary entry) near the descending duodenum **2** to access the Sphincter of Oddi Complex, which is defined by the intersection of the Sphincter Papillae **12**, Sphincter Ampullae **6** and Sphincter Pancreaticus **14**. For example, referring to FIG. **1**, a distal portion of a medical device **200** (e.g., duodenoscope) may be positioned within the descending duodenum, and a guidewire **122** and elongate member **110** advanced through a working channel of the duodenoscope into the major papilla **4** and through the Sphincter Papillae **12**, Sphincter Ampullae **6** and Sphincter Choledochus **8** to access the CBD **10**. Alternatively, the guidewire and endoscopic accessory tool may be advanced through the major papilla **4** and through the Sphincter Papillae **12** and Sphincter Pancreaticus **14** to access the PD **16**. The step of accessing the Sphincter Papillae is notoriously difficult because the opening to the major papilla is extremely small, or even completely collapsed/closed, and extends into the descending duodenum at an angle that is difficult to visualize. Thus, the medical professional is required to manipulate the endoscopic accessory tool and guidewire by manually rotating the duodenoscope and/or using an elevator within the distal head of the duodenoscope. Difficult cannulation procedures in which the medical professional fails to access the Sphincter Papillae within a certain time limit, or after a certain number of unsuccessful attempts, may lead to significant post-procedure complications, such as post-ECRP pancreatitis (PEP).

(15) In various embodiments, the present disclosure relates to medical devices, systems and methods for selective cannulation of the common CBP or PD to treat a variety of hepatobiliary indications during an ECRP procedure. Atraumatic entry of the guidewire and endoscopic accessory tool into the luminal structures of the hepatobiliary system may be achieved through controlled reduction of friction by transmitting mechanical motion to the distal end of an endoscopic accessory tool, and by proxy the guidewire itself.

(16) In embodiments, devices, systems, and methods may transmit motion to the distal tip of an endoscope assembly as described herein and in co-pending application filed concurrently, entitled “Medical Devices, Systems and Methods with Tip Deflection” to Reid et al. (U.S. patent application Ser. No. 16/242,438), which is herein incorporated by reference in its entirety.

(17) Referring to FIGS. **2A-2C**, in one embodiment, an endoscopic accessory tool **100** (e.g., sphincterotome, cannula, catheter, etc.) of the present disclosure may include an elongate member **110** forming a lumen **118** and comprising a proximal end **112**, a distal end **114**, a length and a diameter. The lumen **118** may be configured to receive a guidewire **122** such that a distal portion **124** of the guidewire **122** extends distally beyond the distal end **114** of the elongate member **110**. The elongate member **110** may include a flexible distal portion **116** configured to move (e.g., flex, bend, rotate, wobble, spin, etc.) in a plurality of directions (e.g., x, y and/or z directions), thereby imparting an identical or similar direction of movement to the distal portion **124** of the guidewire **122**. Referring to FIG. **2A**, in one embodiment, the distal portion **116** of the elongate member **110** may be configured to move in a linear motion (e.g., moving in a single direction along a straight line relative to a longitudinal axis of the elongate member **110**), or in a reciprocating motion (e.g., backwards and forwards in a straight line along a longitudinal axis of the elongate member **110**). Referring to FIG. **2B**, in another embodiment, the distal portion **116** of the elongate member **110** may be configured to swing from side-to-side in a vibrating or oscillatory motion. Referring to

FIG. 2C, in another embodiment, the distal portion **116** of the elongate member **110** may be configured to move in a rotary motion (e.g., 360 degrees of rotation around a central axis of the elongate member **110**). As disclosed herein, the medical professional may toggle between any of these motions on demand across a range of frequencies, amplitudes and durations (e.g., brief or sustained pulses) to provide fine-tuned control of the distal portions **116**, **124** of the elongate member **110** and guidewire **122** to facilitate enhanced access into and through patient-specific pathologies (e.g., tortuous anatomies, benign or malignant strictures, blockages due to stones, etc.) that may otherwise inhibit the physical movement of the endoscopic accessory tool into and through the CBD or PD.

(18) As illustrated in FIG. 2B, in one embodiment, the endoscopic accessory tool **100** may further include a cutting wire **120** extending through or along a length of the elongate member **110**. A distal end of the cutting wire **120** may be attached to the elongate member **110** at or near the distal end **114**, and a proximal end of the cutting wire **120** may be attached to an energy source (not shown). A portion of the cutting wire **120** that is coextensive with the distal portion **116** of the elongate member **110** may be unattached and configured to extend at an outward angle relative to a longitudinal axis of the elongate member **110** as the distal portion **116** bends or flexes. With the distal portion **116** of the elongate member **110** held in a bent or flexed position, the medical professional may activate the energy source to energize the angled portion of the cutting wire **120**, thereby providing cutting and cauterizing energy to enlarge an opening of the target body passageway. In various embodiments, the distal portion **116** of the elongate member **110** and/or the distal end of the cutting wire **120** may be configured to deliver real-time feedback information to a medical professional or computer processor, to manually or automatically adjust, e.g., increase or decrease, the frequency, direction and/or duration of movement of the distal portion **116** of the elongate member **110**.

(19) Although the endoscopic accessory tool **100** of the present disclosure may be used as a stand-alone device during an endoscopic procedure, in various embodiments the endoscopic accessory tool **100** may be configured to extend through the working channel of a medical device, e.g., a duodenoscope. Referring to FIGS. 3A-3D, in one embodiment, a medical device **200** of the present disclosure may include an elongate tubular body **130** forming a first lumen **138** (e.g., tool channel, accessory channel, etc.) and comprising a proximal end (not shown), distal end **132**, a length and a diameter. A distal portion **136** of the elongate tubular body **130** may include an opening **134** (e.g., side-opening, etc.) in communication with the first lumen **138**. An elevator **140** (e.g., elevator mechanism, ramp, lift, etc.) may be disposed at a first location within the distal portion **136** of the elongate tubular body **130** (e.g., at junction/intersection of the opening **134** and first lumen **138**). The elevator **140** may be configured to move between a first (e.g., lowered, flat) position and a second (e.g., raised, angled) position relative to a longitudinal axis of the elongate tubular body **130**. In one embodiment, an elevator wire **142** may extend through or along a length of the elongate tubular body **130**, with a distal end **144** of the elevator wire **142** attached to the elevator **140** such that proximally retracting a proximal end (not shown) of the elevator wire **142** may move the elevator **140** from the first position to the second position, and distally advancing the elevator wire **142** may move the elevator **140** from the second position to the first position. The present disclosure is in no way limited to elevators actuated by an elevator wire, but may include a variety of actuation mechanism as are commonly known in the art, including, but not limited to wires, pulleys, pneumatic actuators, wireless or remote-control actuators, and the like.

(20) In various embodiments, the first lumen **138** of the elongate tubular body **130** may be configured to slidably receive an endoscopic accessory tool **100** of the present disclosure. As will be understood by one of skill in the art, a medical professional may manipulate and/or control the distal portions **116**, **124** of the elongate member **110** and guidewire **122** extending through the opening **134** of the medical device **200**, e.g., by rotating or twisting the elongate tubular body **130**, proximally retracting and/or distally extending the elongate member **110** within the first lumen **138**

and/or moving the elevator **140** between the first and second positions.

(21) One or more actuators **149** may be disposed at a second location within the distal portion **136** of the elongate tubular body **130**, and configured to transmit mechanical motion to the distal portion **116** of the elongate member **110** within the distal portion **136** of the elongate tubular body **130**, as discussed below.

(22) Referring to FIG. 3A, in one embodiment, the second location may be adjacent to the first location such that the actuator(s) **149** are configured to transmit mechanical motion in the $-y$ to $+y$ direction through the elevator **140** to the distal portion **116** of the elongate member **110** in contact with the elevator **140**. For example, the actuator(s) **149** may be mounted adjacent to (e.g., in the vicinity of, below, behind, underneath, etc.) or in direct contact with a surface of the elevator **140**. In addition, or alternatively, the actuator(s) **149** may be embedded on or within a portion of the elevator **140**. Referring to FIG. 3B, in another embodiment, the medical device **200** may include a second lumen **139** (e.g., working channel, biopsy channel, etc.) coextensive with the first lumen **138** along the length of the elongate tubular body **130**. The second location may be within or adjacent to a distal end of the second lumen **139** and proximal to the elevator **140** such that the actuator(s) **149** transmit mechanical motion in the $-x$ to $+x$ direction (e.g., back-and-forth) to the distal portion **116** of the elongate member **110** below the opening **134**.

(23) Referring to FIG. 3C, in another embodiment, the second location may be on opposite sides of the first lumen **138** such that the actuator(s) **149** transmit mechanical motion in the $-z$ to $+z$ direction to the elongate member **110**, e.g., proximal to the elevator **140**. Referring to FIG. 3D, in another embodiment, the second location may be on one side (e.g., top side) of the first lumen **138** such that the actuator(s) **149** transmit mechanical motion in the $-y$ to $+y$ direction, e.g., proximal to the portion of the elongate member **110** in contact with the elevator **140**. The number, location, arrangement and/or orientation of actuator(s) **149** is not limited to the embodiments described herein. In various embodiments, any combination of the actuator configurations disclosed herein may be combined to transmit mechanical motion to the elongate member **110** and guidewire **122** in any of the x , y , and/or z directions.

(24) Referring to FIG. 4, in one embodiment, a medical device **300** of the present disclosure may include the identical, or similar, configurations and elements as the medical device **200**, with the exception that the actuators **149** are positioned within an endcap **150** (e.g., endoscopic accessory cap) disposed around an outer surface of the distal portion **136** of the elongate tubular body **130**. The endcap **150** may include an opening **154** configured to align with the opening **134** within the distal portion **136** of the elongate tubular body **130**. In various embodiments, an inner dimension of the opening **154** may be sufficiently large to prevent blocking, occluding or otherwise impairing the function of any accessories, tools, ports, channels or electronics (e.g., light source, camera, etc.) on, within or extending through the distal portion **136** of the elongate tubular body **130**. In one embodiment, the endcap **150** may comprise a variety of polymeric, hard plastic, soft plastic, rubber and/or neoprene materials with an atraumatic outer dimension configured to navigate through tortuous, narrow or otherwise restricted body passageways. The endcap **150** may be permanently attached to the distal portion **136** of the elongate tubular body **130** by a suitable weld, solder, braze, adhesive, epoxy, glue and/or resin. In addition, or alternatively, the endcap **150** may be reversibly attached to the distal portion **136** of the elongate tubular body **130** by at least one of a snap-fit, press-fit, interference fit, compressive fit, static friction fit, thread and collar fit, lock-and-key fit, luer-lock, and the like.

(25) One or more actuators **149** may be disposed on or within the endcap **150** around or adjacent to the opening **154**. As above, the actuator(s) **149** may be arranged in a variety of numbers, locations, arrangements and/or orientations configured to transmit mechanical motion to the distal portion **116** of the elongate member **110** and guidewire **122** extending through the opening **154** in the x , y and/or z directions, e.g., along, across or above the plane of the medical device **300**. In various embodiments, mechanical motion transmitted to a distal portion **116** of the elongate member **110**

extending above the plane of the outer surface of the medical device **300**, e.g., closer to the distal end of the elongate member **110** and guidewire **122**, may provide the medical professional with enhanced control of the endoscopic accessory tool and guidewire when attempting to access the major papilla, steer towards a target duct and/or navigate beyond strictures, obstructions and the like.

(26) Although the medical devices **200**, **300** of the present disclosure include one or more actuators configured to transmit mechanical motion to the distal tip of an elongate member **110** and guidewire **122** of an endoscopic accessory tool **100**, in various embodiments of the present disclosure, one or more actuators may be disposed on or within the endoscopic accessory tool itself.

(27) Referring to FIGS. 5A-5B, in one embodiment, an endoscopic accessory tool **100** of the present disclosure, e.g., as disclosed in FIGS. 2A-2C, may further include one or more actuators **126** attached to an outer surface of the elongate member **110** at or near the distal end **114** by a support or attachment member **128**. In various embodiments, positioning the actuator(s) **126** on the outer surface of the elongate member **110** may prevent the lumen **118** from being narrowed or otherwise occluded. Referring to FIGS. 5C-5D, in another embodiment, the endoscopic accessory tool **100** may further include one or more actuators **126** embedded or integrated within a wall at or near the distal end **114** of the elongate member **110**. In any of the embodiments depicted in FIGS. 5A-5D, the actuator(s) **126** may be connected to a power supply (not shown) by an electrically conductive wire **129** extending along, or embedded within, an inner or outer surface along a length of the elongate member **110**. In various embodiments, the wire **129** may be attached to the actuators **126** using a suitable biocompatible adhesive, glue, resin or shrink wrap material, etc. In addition, or alternatively, an entire length or intermittent portions of the conductive wire **129** may be attached to an inner or outer surface of the elongate member **110** as discussed above. Although FIGS. 5A-5D depict two actuators **126** disposed on opposite sides of the distal end **114** of the elongate member **110**, in various embodiments, the number, location, arrangement, orientation and/or size of actuators **126** may vary. For example, the actuators **126** may be arranged in an asymmetric pattern to urge or bias the distal portion **116** of the elongate member **110** and guidewire (not shown) to vibrate and move in a particular direction. In addition, or alternatively, opposing actuators **126** with different amplitudes of vibration may be configured to urge or bias the distal portion **116** of the elongate member **110** and guidewire (not shown) to vibrate and move in a particular direction, e.g., vibration is biased in the direction of the actuator(s) with the greater amplitude of vibration. In various embodiments, the ability to transmit mechanical motion directly within the distal end **114** of the elongate member **110** may provide the medical professional with finer control of the endoscopic accessory tool **100** when accessing the ampullary entry. In addition, or alternatively, the ability to preferentially urge or bias the direction of vibration, as discussed above, may allow the medical professional to more precisely steer with guidewire **122** and elongate member **110** into the CBD or PD, e.g., as the asymmetric vibration is intermittently toggled on/off.

(28) Referring to FIGS. 6A-6B, in one embodiment, an endoscopic accessory tool **100** of the present disclosure, e.g., as disclosed in FIGS. 2A-2C, may further include one or more actuators **126** disposed within a proximal portion of the elongate member **110**, e.g., within or near a handle of the endoscopic accessory tool **100**, to propagate vibrational energy along a full length of the elongate member **110** and transmit mechanical motion to the distal portion **116** of the elongate member **110** and guidewire **122**, as discussed above. Without limiting the present disclosure to specific compositions or compounds, in one embodiment, the endoscopic accessory tool **100** may include a metallic material (e.g., rod, shaft, beam, wire, coil, etc.) which spans the full length of the elongate member **110** (e.g., along an inner or outer surface of the elongate member, or embedded within the wall of the elongate member) to vibrationally connect the actuator(s) **126** to the distal portion **116** of the elongate member **110** and guidewire **122**. In various embodiments, mechanical motion transmitted in the $-z$ to $+z$ direction within the proximal portion of the elongate member

110 may urge the distal portion **124** of the guidewire **122** to move linearly in the $-x$ to $+x$ direction. In one embodiment, in addition to propagating vibrational energy along the full length of the elongate member **110**, the actuators(s) **126** may be configured to transmit mechanical motion directly onto, and along a length of, the guidewire **122** (FIG. 6A) and/or the cutting wire **120** (FIG. 6B) extending along or through the elongate member **110**.

(29) The actuators **149**, **126** of any of the medical devices **200**, **300** or endoscopic accessory tools **100** of the present disclosure may include acoustic, electromechanical and/or piezoelectric elements, as are commonly known in the art, to transmit mechanical motion to a distal portion of the endoscopic accessory tool. Electrical power may be delivered to these acoustic, electromechanical and/or piezoelectric elements using, by way of non-limiting example, batteries (e.g., external to or as a component of the medical device or endoscopic accessory tool), wires or other electrically conductive circuits or materials extending along or through a length of the medical device or endoscopic accessory tool, or plug-and-play adaptors integrated with a serial port (e.g., Bluetooth, RFID chips, sensors, etc.) located on or within a distal portion of the medical device or endoscopic accessory tool. Using a step pedal, push button, joystick, touch screen, voice command, and the like, the medical professional may transfer power (e.g., toggle on/off, increase or decrease the amplitude and/or direction of vibration) to any or all of the actuators on demand with real-time control while imaging the target passageway, e.g., through an endoscopic camera or using external fluoroscopic or radiologic guidance.

(30) Endoscopic accessory tools of the present disclosure are not limited to the embodiments of FIGS. 5A-5D and 6A-6B, e.g., which may utilize acoustic, electromechanical and/or piezoelectric actuators to transmit mechanical motion to the distal portion **116** of the elongate member **110**. Referring to FIGS. 7A-7D, in various embodiments, an endoscopic accessory tool **100** of the present disclosure may include an elongate member **110** and guidewire **122** with corresponding surface features configured to transmit mechanical motion to the distal portion **116** of the elongate member **110** and, by proxy the guidewire **122** itself, as the surface features move relative to each other. Referring to FIG. 7A, in one embodiment, an inner surface **162** of the distal portion **116** of the elongate member **110** may include a series of surface features **164** (e.g., gear-rack, grooves, etc.) configured to mate with a corresponding series of surface features **174** (e.g., ratchet, teeth, etc.) on an outer surface **172** of a distal portion **124** of the guidewire **122**. As the elongate member **110** and guidewire **122** move relative to each other, e.g., by proximally retracting and distally advancing the guidewire **122** within the lumen **118** of the elongate member **110** (e.g., back-and-forth in the $-x$ to $+x$ direction), the surface features **164**, **174** may move across each other to transmit mechanical motion to the distal portions **116**, **124** of the elongate member **110** and guidewire **122**. Referring to FIG. 7B, in one embodiment, the surface features **164**, **174** may be disposed on one side of the elongate member **110** and guidewire **122** to transmit unidirectional mechanical motion to the distal portions **116**, **124** of the elongate member **110** and guidewire **122**. Alternatively, referring to FIG. 7C, in one embodiment, the surface features **164**, **174** may be disposed around an entire circumference of the elongate member **110** and guidewire **122** to transmit random or multi-directional mechanical motion to the distal portions **116**, **124** of the elongate member **110** and guidewire **122**. Referring to FIG. 7D, in one embodiment, an inner surface **162** of the proximal end **112** of the elongate member **110** may include a series of surface features **164** configured to mate with a corresponding series of surface features **174** on an outer surface **172** of a proximal end of the guidewire **122**. As the elongate member **110** and guidewire **122** move relative to each other, e.g., by proximally retracting and distally advancing the guidewire **122** within the lumen **118** of the elongate member **110** (e.g., back-and-forth in the $-x$ to $+x$ direction), the surface features **164**, **174** may move across each other to propagate vibrational energy along a full length of the elongate member **110** and guidewire **122** to transmit mechanical motion to the distal portion **116** of the elongate member **110** and guidewire **122**, as discussed above. In various embodiments, the medical professional may move the elongate member **110** and guidewire relative to each other

using, e.g., a push-mechanism or roller mechanism mounted within a handle of the endoscopic accessory tool **100**.

(31) Referring to FIGS. **8A-8D**, in various embodiments, an endoscopic accessory tool **100** of the present disclosure may include an elongate member **110** and guidewire **122** with corresponding surface features configured to transmit mechanical motion to a full length of the guidewire **122** as the elongate member **110** is rotated, e.g., an alternating side-to-side or circular/spinning direction. Referring to FIG. **8A**, in one embodiment, an inner surface **162** of the distal portion **116** of the elongate member **110** may include a surface feature **166** (e.g., notch, slot, etc.) configured to mate with a corresponding surface feature **176** (e.g., bump, protrusion, etc.) on an outer surface **172** of a distal portion **124** of the guidewire **122**. With the surface features **166**, **176** of the elongate member **110** and guidewire **122** engaged (FIG. **8B**), a proximal end of the elongate member **110** may be rotated as describe above to impart corresponding rotation along a full length of the guidewire **122**. In various embodiments, a distal portion **124** of the guidewire **122** extending distally beyond the distal end **114** of the elongate member **110** may bend, flop, twist and/or rotate in a variety of directions as the interlocked elongate member **110** and guidewire **122** are rotated. In various embodiments, the interlocking surface features are not limited to the notch and protrusion configuration described above, but may include corresponding teeth and groove configurations (FIG. **8C**) or corresponding sinusoidal or wave-form configurations (FIG. **8D**).

(32) The medical devices of the present disclosure are not limited to duodenoscopes, and may include a variety of medical devices for accessing body passageways, including, for example, catheters, ureteroscopes, bronchoscopes, colonoscopes, arthroscopes, cystoscopes, hysteroscopes, and the like.

(33) All of the devices and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the devices and methods of this disclosure have been described in terms of preferred embodiments, it may be apparent to those of skill in the art that variations can be applied to the devices and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the disclosure. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the disclosure as defined by the appended claims.

Claims

1. A medical device, comprising: an elongate tubular body having a proximal end, a distal end, and a length therebetween, the elongate tubular body including a first lumen, wherein a distal portion of the elongate tubular body includes a side opening in communication with the first lumen, wherein the medical device is configured to receive an elongate member in the first lumen of the elongate tubular body and to direct a distal end of the elongate member through the side opening of the elongate tubular body; and an actuator disposed within the elongate tubular body, the actuator configured to transmit mechanical motion to the distal portion of the elongate member when positioned within the distal portion of the elongate tubular body.
2. The medical device of claim 1, wherein the actuator is disposed within the distal portion of the elongate tubular body.
3. The medical device of claim 1, wherein the medical device comprises an elevator disposed within the distal portion of the elongate tubular body, the elevator configured to direct the distal end of the elongate member through the side opening.
4. The medical device of claim 3, wherein the actuator is configured to transmit mechanical motion to the elevator.
5. The medical device of claim 1, wherein the medical device comprises a ramp disposed within the distal portion of the elongate tubular body, the ramp configured to direct the distal end of the

elongate member through the side opening.

6. The medical device of claim 5, wherein the actuator is configured to transmit mechanical motion to the ramp.
 7. The medical device of claim 1, wherein the medical device is a duodenoscope.
 8. A system comprising: the medical device of claim 1; and the elongate member.
 9. The system of claim 8, wherein the elongate member is an endoscopic accessory tool.
 10. The system of claim 8, further comprising a guidewire.
 11. The system of claim 8, wherein the medical device is a duodenoscope.
 12. The system of claim 8, wherein the actuator is disposed within the distal portion of the elongate tubular body.
 13. The system of claim 8, wherein the medical device comprises an elevator disposed within the distal portion of the elongate tubular body, the elevator configured to direct the distal end of the elongate member through the side opening.
 14. The system of claim 13, wherein the actuator is configured to transmit mechanical motion to the elevator.
 15. The system of claim 8, wherein the medical device comprises a ramp disposed within the distal portion of the elongate tubular body, the ramp configured to direct the distal end of the elongate member through the side opening.
 16. The system of claim 15, wherein the actuator is configured to transmit mechanical motion to the ramp.
 17. A method comprising: inserting into a first body passageway of a patient the elongate tubular body of the medical device of claim 1, the elongate tubular body advanced such that the side opening is positioned proximate an opening of a second body passageway that intersects with the first body passageway; advancing the distal end of the elongate member through the first device lumen and the side opening of the elongate tubular body; and energizing the actuator, the actuator transmitting the mechanical motion to the distal portion of the elongate member; and accessing the opening in the second body passageway.
 18. The method of claim 17, wherein the elongate member is an endoscopic accessory tool.
 19. The method of claim 18, wherein the first body passageway comprises a descending duodenum, the second body passageway comprises a major papilla of the duodenum, and wherein the endoscopic accessory tool is advanced through the major papilla.
 20. The method of claim 19, wherein the endoscopic accessory tool and a guidewire are advanced through the major papilla.
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